Mayank Singh

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INTERESTS

Few-shot and Self-supervised Learning, 3D Computer Vision, Generative Models

EDUCATION

CARNEGIE MELLON UNIVERSITY
School of Computer Science
Master of Science in Robotics
Jan. 2021 - Dec 2022
Pittsburgh, PA
GPA: 4.05/4.3

IIT KHARAGPUR
B.Sc/M.Sc in Mathematics & Computing
2012-2017 | Kharagpur, India
GPA: 8.85 / 10.0

COURSEWORK

GRADUATE
Machine learning, Computer Vision, Reinforcement Learning

UNDERGRADUATE
Linear Algebra
Data Structures and Algorithms

ACHIEVEMENTS

YOUNG DATA SCIENTIST AWARD
Won the data science coding talent competition at ZS Associate, 2017

SKILLS

Proficient in : Python • PyTorch • Numpy • Scikit-learn
Comfortable in : C++ • Apache Spark • TensorFlow

POSITIONS

Teaching Assistant:
Regression & Time Series (IIT KGP),
Machine Learning (Adobe)

EXPERIENCE

APPLE | Machine Learning Engineer
13 Feb 2023 – Present | Sunnyvale, CA
- Working on 3D computer vision algorithms for Augmented reality applications.

CMU | Research Assistant under Prof. Katerina Fragkiadaki
Feb 2021 – Dec 2022 | Pittsburgh, PA
- Worked on 3D object detection/reconstruction and few-shot learning.

APPLE | Intern
16 May 2022 – 26 Aug 2022 | Sunnyvale, CA
- Worked on generative models for 3D computer vision.

ADOBE | Member of Technical Staff 2
18 July 2017 – 29 Jan 2021 | Noida, India
- Developed a Video Tutorial Recommendation system for Adobe Creative Cloud.
- Designed a reinforcement learning method to personalize user experience.

ADOBE | Intern
May 2016 – July 2016 | Noida, India
- Developed a novel recommendation system by jointly modeling items and users.

PUBLICATIONS

ANALOGY-FORMING TRANSFORMERS FOR FEW-SHOT 3D PARSING **| ICLR 2023 (paper link)**
- Presented a novel architecture Analogical Networks, that casts fine-grained 3D visual parsing as analogy-forming inference.
- Improved state-of-the-art 3D segmentation in many-shot, meta-learning and few-shot setting.

ATTRIBUTIONAL ROBUSTNESS TRAINING USING INPUT-GRADIENT SPATIAL ALIGNMENT **| ECCV 2020 (paper link)**
- Achieved state-of-the-art attributional and adversarial robustness by a margin of $\approx 6 - 18\%$ on standard computer vision datasets.

DATA INSTANCE PRIOR (DISP) IN GENERATIVE ADVERSARIAL NETWORKS **| WACV, 2022 (paper link)**
- Proposed a transfer learning method for GAN training in low data setting that improved image generation quality in limited data setting.

CHARTING THE RIGHT MANIFOLD: MANIFOLD MIXUP FOR FEW-SHOT LEARNING **| (Spotlight Presentation) MetaLearn workshop, NeurIPS 2019 | WACV 2020 (paper link)**
- Analyzed the role of self-supervision techniques along with Manifold-Mixup augmentation in few-shot image classification.
- Improved the state-of-the-art few-shot accuracy on mini-ImageNet, CUB and tiered-ImageNet datasets by $\approx 3 - 8\%$.

LT-GAN: SELF-SUPERVISED GAN WITH LATENT TRANSFORMATION DETECTION **| WACV, 2021 (paper link)**
- Proposed a self-supervised approach to improve the generation quality of images by estimating the GAN-induced transformation.

PATENTS

- Machine-learning based multi-step engagement strategy generation and visualization (US10609434B2)
- Generating trained neural networks with increased robustness against adversarial attacks (US20200234110A1)